

CHAMBERS'S JOURNAL

OF

POPULAR

LITERATURE, SCIENCE, AND ART

Fifth Series

ESTABLISHED BY WILLIAM AND ROBERT CHAMBERS, 1832

CONDUCTED BY R. CHAMBERS (SECUNDUS)

No. 40.—VOL. I.

SATURDAY, OCTOBER 4, 1884.

PRICE 1½d.

THE FUTURE OF ELECTRICITY AND GAS.

MORE than eighty years ago, Davy first produced and exhibited the arc-light to an admiring and dazzled audience at the Royal Institution; and forty years later, at the same place, Faraday, by means of his memorable experiments in electro-dynamics, laid down the laws on which the modern dynamo-electric machine is founded. Though known at the beginning of the century, the electric light remained little more than a scientific curiosity until within the last ten years, during which period the dynamo-electric machine has been brought to its present perfection, and electric lighting on a large and economical scale thus rendered possible. The first practical incandescent lamps were produced only seven years ago, though the idea of lighting by incandescence dates back some forty years or more; but all attempts to manufacture an efficient lamp were rendered futile by the impossibility of obtaining a perfect vacuum. The year 1881 will long be remembered as that in which electric lighting by incandescence was first shown to be possible and practicable.

The future history of the world will doubtless be founded more or less on the history of scientific progress. No branch of science at present rivals in interest that of electricity, and at no time in the history of the world has any branch of science made so great or so rapid progress as electrical science during the past five years.

And now it may be asked, where are the evidences of this wonderful progress, at least in that branch of electricity which is the subject of the present paper? Quite recently, the wonders of the electric light were in the mouths of every one; while at present, little or nothing is heard about it except in professional quarters. Is the electric light a failure, and are all the hopes that have been placed on it to end in nothing? Assuredly not. The explanation of the present lull in electric lighting is not far to

seek; it is due almost solely and entirely to speculation. The reins, so to say, had been taken from the hands of engineers and men of science; the stock-jobbers had mounted the chariot, and the mad gallop that followed has ended in ruin and collapse. Many will remember the electric-light mania several years ago, and the panic that took place among those holding gas shares. The public knew little or nothing about electricity, and consequently nothing was too startling or too ridiculous to be believed. Then came a time of wild excitement and reckless speculation, inevitably followed by a time of depression and ruination. Commercial enterprise was brought to a stand-still; real investors lost all confidence; capital was diverted elsewhere; the innocent suffered, and are still suffering; and the electric light suffered all the blame. The government was forced to step in for the protection of the public; and the result of their legislation is the Electric Lighting Act, which authorises the Board of Trade to grant licenses to Companies and local authorities to supply electricity under certain conditions. These conditions have reference chiefly to the limits of compulsory and permissive supply, the securing of a regular and efficient supply, the safety of the public, the limitation of prices to be charged, and regulations as to inspection and inquiry.

That the electric light has not proved a failure may be gleaned from a rough survey of what has been done during the past two years, in spite of unmerited depression and depreciation. In this country, permanent installations have been established at several theatres in London and the provinces; the Royal Courts of Justice, the Houses of Parliament, Buckingham Palace, Windsor Castle, the Bank of England, and other well-known buildings; while numerous railway stations, hotels, clubs, factories, and private mansions throughout the country, have also adopted the new light either entirely or in part. In addition to this, over forty steamships have been fitted with the electric light during the past year; and the Holborn Viaduct, with its shops

CHAMBERS'S JOURNAL

OF
POPULAR

LITERATURE, SCIENCE, AND ART

Fifth Series

ESTABLISHED BY WILLIAM AND ROBERT CHAMBERS, 1832

CONDUCTED BY R. CHAMBERS (SECUNDUS)

No. 40.—VOL. I.

SATURDAY, OCTOBER 4, 1884.

PRICE 1½d.

THE FUTURE OF ELECTRICITY AND GAS.

MORE than eighty years ago, Davy first produced and exhibited the arc-light to an admiring and dazzled audience at the Royal Institution; and forty years later, at the same place, Faraday, by means of his memorable experiments in electro-dynamics, laid down the laws on which the modern dynamo-electric machine is founded. Though known at the beginning of the century, the electric light remained little more than a scientific curiosity until within the last ten years, during which period the dynamo-electric machine has been brought to its present perfection, and electric lighting on a large and economical scale thus rendered possible. The first practical incandescent lamps were produced only seven years ago, though the idea of lighting by incandescence dates back some forty years or more; but all attempts to manufacture an efficient lamp were rendered futile by the impossibility of obtaining a perfect vacuum. The year 1881 will long be remembered as that in which electric lighting by incandescence was first shown to be possible and practicable.

The future history of the world will doubtless be founded more or less on the history of scientific progress. No branch of science at present rivals in interest that of electricity, and at no time in the history of the world has any branch of science made so great or so rapid progress as electrical science during the past five years.

And now it may be asked, where are the evidences of this wonderful progress, at least in that branch of electricity which is the subject of the present paper? Quite recently, the wonders of the electric light were in the mouths of every one; while at present, little or nothing is heard about it except in professional quarters. Is the electric light a failure, and are all the hopes that have been placed on it to end in nothing? Assuredly not. The explanation of the present lull in electric lighting is not far to

seek; it is due almost solely and entirely to speculation. The reins, so to say, had been taken from the hands of engineers and men of science; the stock-jobbers had mounted the chariot, and the mad gallop that followed has ended in ruin and collapse. Many will remember the electric-light mania several years ago, and the panic that took place among those holding gas shares. The public knew little or nothing about electricity, and consequently nothing was too startling or too ridiculous to be believed. Then came a time of wild excitement and reckless speculation, inevitably followed by a time of depression and ruin. Commercial enterprise was brought to a stand-still; real investors lost all confidence; capital was diverted elsewhere; the innocent suffered, and are still suffering; and the electric light suffered all the blame. The government was forced to step in for the protection of the public; and the result of their legislation is the Electric Lighting Act, which authorises the Board of Trade to grant licenses to Companies and local authorities to supply electricity under certain conditions. These conditions have reference chiefly to the limits of compulsory and permissive supply, the securing of a regular and efficient supply, the safety of the public, the limitation of prices to be charged, and regulations as to inspection and inquiry.

That the electric light has not proved a failure may be gleaned from a rough survey of what has been done during the past two years, in spite of unmerited depression and depreciation. In this country, permanent installations have been established at several theatres in London and the provinces; the Royal Courts of Justice, the Houses of Parliament, Buckingham Palace, Windsor Castle, the Bank of England, and other well-known buildings; while numerous railway stations, hotels, clubs, factories, and private mansions throughout the country, have also adopted the new light either entirely or in part. In addition to this, over forty steamships have been fitted with the electric light during the past year; and the Holborn Viaduct, with its shops

**The Project Gutenberg eBook of Chambers's Journal of
Popular Literature, Science, and Art, Fifth Series, No. 40, Vol.
I, October 4, 1884**

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: Chambers's Journal of Popular Literature, Science, and Art, Fifth Series, No. 40, Vol. I, October 4, 1884

Author: Various

Release date: September 29, 2021 [eBook #66418]

Most recently updated: October 18, 2024

Language: English

Other information and formats: www.gutenberg.org/ebooks/66418

Credits: Susan Skinner, Eric Hutton and the Online Distributed Proofreading Team at <https://www.pgdp.net> (This file was produced from images generously made available by The Internet Archive)

*** START OF THE PROJECT GUTENBERG EBOOK CHAMBERS'S
JOURNAL OF POPULAR LITERATURE, SCIENCE, AND ART, FIFTH
SERIES, NO. 40, VOL. I, OCTOBER 4, 1884 ***

**CHAMBERS'S JOURNAL
OF
POPULAR
LITERATURE, SCIENCE, AND
ART.**

CONTENTS

THE FUTURE OF ELECTRICITY AND GAS.

BY MEAD AND STREAM.

VÆ VICTIS!

ONE WOMAN'S HISTORY.

POPULAR AMUSEMENTS IN GERMANY.

PRINTERS' ERRORS.

'THE KING COUNTRY.'

THE MOULMEIN ELEPHANTS.

GUM-ARABIC AND THE SOUDAN.

ONE BY ONE.



No. 40.—VOL. I.

SATURDAY, OCTOBER 4, 1884.

PRICE 1½d.

THE FUTURE OF ELECTRICITY AND GAS.

MORE than eighty years ago, Davy first produced and exhibited the arc-light to an admiring and dazzled audience at the Royal Institution; and forty years later, at the same place, Faraday, by means of his memorable experiments in electro-dynamics, laid down the laws on which the modern dynamo-electric machine is founded. Though known at the beginning of the century, the electric light remained little more than a scientific curiosity until within the last ten years, during which period the dynamo-electric machine has been brought to its present perfection, and electric lighting on a large and economical scale thus rendered possible. The first practical incandescent lamps were produced only seven years ago, though the idea of lighting by incandescence dates back some forty years or more; but all attempts to manufacture an efficient lamp were rendered futile by the impossibility of obtaining a perfect vacuum. The year 1881 will long be remembered as that in which electric lighting by incandescence was first shown to be possible and practicable.

The future history of the world will doubtless be founded more or less on the history of scientific progress. No branch of science at present rivals in interest that of electricity, and at no time in the history of the world has any branch of science made so great or so rapid progress as electrical science during the past five years.

And now it may be asked, where are the evidences of this wonderful progress, at least in that branch of electricity which is the subject of the present paper? Quite recently, the wonders of the electric light were in the mouths of every one; while at present, little or nothing is heard about it except in professional quarters. Is the electric light a failure, and are all the hopes that have been placed on it to end in nothing? Assuredly not. The explanation of the present lull in electric lighting is not far to seek; it is due almost solely and entirely to speculation. The reins, so to say, had been taken from the hands of engineers and men of science; the stock-jobbers had mounted the chariot, and the mad gallop that followed has ended in ruin and collapse. Many will remember the electric-light mania several years

ago, and the panic that took place among those holding gas shares. The public knew little or nothing about electricity, and consequently nothing was too startling or too ridiculous to be believed. Then came a time of wild excitement and reckless speculation, inevitably followed by a time of depression and ruination. Commercial enterprise was brought to a standstill; real investors lost all confidence; capital was diverted elsewhere; the innocent suffered, and are still suffering; and the electric light suffered all the blame. The government was forced to step in for the protection of the public; and the result of their legislation is the Electric Lighting Act, which authorises the Board of Trade to grant licenses to Companies and local authorities to supply electricity under certain conditions. These conditions have reference chiefly to the limits of compulsory and permissive supply, the securing of a regular and efficient supply, the safety of the public, the limitation of prices to be charged, and regulations as to inspection and inquiry.

That the electric light has not proved a failure may be gleaned from a rough survey of what has been done during the past two years, in spite of unmerited depression and depreciation. In this country, permanent installations have been established at several theatres in London and the provinces; the Royal Courts of Justice, the Houses of Parliament, Buckingham Palace, Windsor Castle, the Bank of England, and other well-known buildings; while numerous railway stations, hotels, clubs, factories, and private mansions throughout the country, have also adopted the new light either entirely or in part. In addition to this, over forty steamships have been fitted with the electric light during the past year; and the Holborn Viaduct, with its shops and buildings, has been lighted without interruption for the past two years. On the continent, in addition to a large number of factories, private houses, and public buildings, numerous theatres at Paris, Munich, Stuttgart, Brunn, Vienna, Berlin, Prague, and Milan have been electrically lighted. In New York, an installation of ten thousand lights has been successfully running for the last year or two. Any one wishing to see the electric light to advantage and its suitability to interior decoration, should visit the Holborn Restaurant. This building, with its finely decorated rooms, its architectural beauties, and ornamental designs in the renaissance style, when viewed by the electric light, is without doubt one of the chief sights of London.

The electric light in the form of the well-known powerful and dazzling arc-light is the favourite illuminant for lighting harbours, railway stations, docks, public works, and other large spaces. But it is to the incandescent lamp that one must look par excellence for the 'light of the future.' It has been satisfactorily established that lighting by incandescence is as cheap as lighting by gas, provided that it be carried out on an extensive scale.

Very contradictory statements have from time to time been published as to the relative cost of lighting by electricity and gas; and a few remarks on the subject, without entering into detailed figures, will explain much of this discrepancy. These remarks will refer to electric lighting by incandescence.

In the first place, the lighting may be effected in one of three ways—(1) by primary batteries; (2) by dynamo-machines; or (3) by a combination of dynamo-machines and secondary batteries. The expense of working with primary batteries is altogether prohibitory, except in the case of very small installations; while secondary batteries have not yet been made a practical success; so that the second method mentioned above is the only one at present in the field. In the second place, a distinction must be made between isolated installations and a general system of lighting from central stations. Up to the present time, nearly all the lighting by electricity has been effected by isolated installations. If every man requiring one hundred or even several hundred lights were to set up his own gas-works and supply himself from them, the cost of lighting by gas would be enormously increased. Hence it is manifestly unfair to compare the cost of electric light obtained from isolated installations with gas obtained from gas-works supplying many thousands of lights; yet this is being constantly done. Central stations supplying at least, say, ten thousand lights, and gas-works on an equal scale, must be compared in order to arrive at a true estimate of the relative cost of electricity and gas. Several such extended installations are now being erected in London and elsewhere. With improved generating apparatus, and above all, with improved lamps, it is confidently anticipated that the electric light will eventually be cheaper than gas. Even if dearer than gas, it will be largely used for lighting dwelling-houses, theatres, concert-halls, museums, libraries, churches, shops, show-rooms, factories, and ships; while perhaps gas may long hold its own as the poor man's friend, since it affords him warmth as well as light.

The incandescent light is entirely free from the products of combustion which heat and vitiate the air; it enables us to see pictures and flowers as by daylight; it supports plants instead of poisoning them, and enables many industries to be carried on by night as well as by day. Add to this an almost perfect immunity from danger of fire and no fear of explosion. When it is realised that a gas flame gives out seventeen times as much heat as an incandescent lamp of equal light-giving power, and that an ordinary gas flame vitiates the air as much as the breathing of ten persons, some idea may be formed of the advantage of the electric light from a sanitary point of view. To this may be added absence of injury to books, walls, and ceilings. Visitors to the Savoy Theatre in London will doubtless have seen the adaptability of this light for places of public amusement, and it is now possible to sit out a play in a cool and pleasant atmosphere without incurring a severe headache. To theatrical managers the light offers in addition unusual facilities for producing spectacular effects, such as the employment of green, red, and white lamps to represent night, morning, and daylight. The freedom from weariness and lassitude after spending an evening in an electrically lighted apartment must be experienced in order to be appreciated. The electric light very readily adapts itself to the interior fittings and decorations of houses and public buildings, and it can be placed in positions where gas could not be used on account of the danger of fire. The old lines of gas-fittings should be avoided as far as possible, and the lights placed singly where required, and not 'bunched' together. For the lighting of mines, electricity must stand unrivalled, though little has as yet been done in this direction. Its speedy adoption either voluntarily or by Act of Parliament, with the employment of lime cartridges instead of blasting by gunpowder, will in the future render explosions in mines almost an impossibility. In some cases, gas may yet for some time compete with the electric light both in brilliancy and economy; for the electric light has spurred on the gas Companies to the improved lighting of many of our public streets and places.

With the general introduction of electricity for the purpose of lighting comes the introduction of electricity for the production of power; for the same current entering by the same conductors can be used for the production of light or of power, or of both. The same plant at the central stations will supply power by day and light by night, with evident economy. Electricity will thus be used for driving sewing-machines, grinding, mixing,

brushing, cleaning, and many other domestic purposes. In many trades requiring the application of power for driving light machinery for short periods, electricity will be of the greatest value, and artisans will have an ever ready source of power at their command in their own homes.

Is electricity to supersede gas altogether? By no means, for gas is destined to play a more important part in the future than it has done in the past. Following close upon the revolution in the production of light comes a revolution in the production of heat for purposes of warming and cooking, and for the production of power. Gas in the future will be largely used not necessarily as an illuminant, but as a fuel and a power producer. When gas is burned in an ordinary gas flame, ninety-five per cent. of the gas is consumed in producing heat, and the remaining five per cent. only in producing light. Gas is far more efficient than raw coal as a heating agent; and it is also far cheaper to turn coal into gas and use the gas in a gas-engine, than to burn the coal directly under the boiler of a steam-engine; for gas-engines are far more economical than steam-engines. Bearing these facts in mind, it cannot but be seen that the time is not far distant when, both by rich and poor, gas will be used as the cheapest, most cleanly, and most convenient means for heating and cooking, and raw coal need not enter our houses; also that gas-engines must sooner or later supersede steam-engines, and gas thus be used for driving the machine that produces the electricity. In the case of towns distant not more than, say, fifty miles from a coal-field, the gas-works could with advantage be placed at the colliery, the gas being conveyed to its destination in pipes. Thus, coal need no longer be seen, except at the colliery and the gas-works. With the substitution of gas for coal, as a fuel, will end the present abominable and wasteful production of smoke. When smoke, 'blacks,' and noxious gases are thus done away with, life in our most populous towns may become a real pleasure. Trees, grass, and flowers will flourish, and architecture be seen in all its beauty. Personal comfort will be greatly enhanced by the absence of smuts, 'pea-soup' fogs, and noxious fumes; and monuments, public buildings, and pictures saved from premature destruction.

The present method of open fires is dirty, troublesome, wasteful, and extravagant. With the introduction of gas as a heating agent, there will be no more carting about of coals and ashes, and no more troublesome lighting of fires with wood, paper, and matches. No more coal-scuttles, no more smoky

chimneys, no more chimney-sweeps! On the other hand, the old open coal-fire is cheerful, 'pokable,' and conducive to ventilation; while the Englishman loves to stand in front of it and toast himself. All this, however, may still be secured in the gas stoves of the future, as any one could easily have satisfied himself at the recent Smoke Abatement Exhibition in London. The gas stove of the future must be an open radiating stove, and not a closed stove, which warms the air by conduction and convection chiefly, and renders the air of a room dry and uncomfortable.

It has frequently been pointed out that our coal-fields are not inexhaustible; but they doubtless contain a sufficient supply for hundreds of years to come. Long before the supply is likely to run short, other sources of nature will be largely drawn upon. These are the winds, waterfalls, tides, and the motion of the waves. The two former have to some extent been utilised; but little or nothing has been done or attempted with the latter. Before these can be to any extent made use of, means must be devised for storing energy in the form of electricity, a problem which is now being vigorously attacked, but as yet without much practical success. That electricity has a great future before it cannot for a moment be doubted.

BY MEAD AND STREAM.

BY CHARLES GIBBON.

CHAPTER L.—A CROW TO PLUCK.

THE two men stared at each other—Mr Hadleigh with an expression of stern inquiry; Caleb with a sullen audacity which failed to conceal the confusion and disappointment he felt. But he made no attempt to apologise, to explain, or to retreat.

After a brief inspection of the man, Mr Hadleigh was reassured: this was no common burglar he had to deal with, and no immediate violence was to be feared.

‘My good man,’ he said calmly, ‘you have wasted your time and labour if you expected to find money or plate here. That safe, which you see is open, contains my cheque-books; but they are worthless to you without my signature. As for what plate and jewels there may be to reward your adventure, they are in different parts of the house, and before you can leave this room to seek them you must murder me. And before you do that, I shall sound this alarm.’

As he spoke he took up a green cord which lay beside his desk. The cord communicated with a bell in the butler’s room, which if rung at that time of night would certainly have aroused the household.

‘I didn’t come here to rob; I didn’t expect to find *you* in this room, and I don’t mean to hurt you.’

Gruff and surly as Caleb’s manner was, Hadleigh, even in that moment of peril, did him justice.

‘I believe you, Kersey,’ he said quietly; ‘and to prove it, I shall sit down and listen to what explanation you have to give. Something very unusual there must be to have caused you to act as you have done. I told you at the end of

the harvest that if I could serve you at any time, I should be pleased to do so. Is that why you have come?’

‘No,’ was the sulky answer.

Although tortured by passion, Caleb was not only sensible of the confidence which Mr Hadleigh showed him under such peculiar circumstances, but felt his self-respect raised by it, and was wishful to make matters clear. The thing somehow stuck in his throat, for he who had broken into the house at midnight had to tell this man of his son’s guilt—as he believed—and of Pansy’s shame.

‘Then what did bring you here and in such an outrageous fashion?’

‘I thought to find your son Mr Coutts here. I’ve been waiting for him nigh on six hours. When he came, he wouldn’t tell me the truth, wouldn’t wait to speak to me, and I am determined that he shall—not only speak to me this night, but speak true. I thought I heard him hollering to me from that window. I didn’t want to make a row if it could be helped, so I got a ladder and came in, meaning to ask him to do things straight and quiet. That’s all.’

As Coutts’s room was above the library, Mr Hadleigh comprehended the mistake Caleb had made, and was satisfied that no intention of robbery had brought him there. His own intense preoccupation had prevented him from observing any disturbance.

‘What is it you wish him to speak truth about?’ he inquired.

‘I’d rather speak to himself,’ was the gloomy response.

‘You are aware that breaking into the house in this way might be an awkward thing for you if brought before a magistrate. But since the matter is important enough to induce you to run the risk you have done to-night, you had better take me into your confidence. I have no doubt of being able to assist you.’

‘Well, then,’ said Caleb, after another minute of hesitation, and the blood tingling in his honest cheeks on her account, ‘I want to know what he has done with Pansy Culver.’

‘What can he have to do with the girl?’

‘More than I care to think—more, maybe, than you would care to learn. He has wiled her away from home and won’t tell me where she is.’

‘There must be some stupid mistake here, Kersey. Mr Coutts Hadleigh is too careful of his reputation to perpetrate such a monstrous act. On what grounds do you accuse him?’

Bluntly and speaking with less difficulty now that the ice was broken, Caleb gave his reasons for believing Coutts guilty—what he had observed at their chance meetings, and particularly her rejection of himself after she had led him to think that she favoured him. Mr Hadleigh allowed him to tell his story to the end without interruption. He could see that the man was blinded by jealousy and rage, was unconsciously exaggerating trifles, and distorting them into proofs of his foregone conclusion.

‘It is fortunate that accident has made me the first hearer of this accusation,’ he said calmly, when Caleb stopped. ‘I had little doubt from the first that you were labouring under a delusion: I am now convinced of it. I will undertake to convince you of it in the morning, if you will be advised and remain quiet to-night.’

‘I won’t wait till morning—I’ll have it out of him now. Where is he?’

‘Stay where you are, sir!’ said Mr Hadleigh authoritatively, as Caleb made a movement towards the door. ‘If you have no care for yourself, you must have some for the girl. A brawl between you and my son on her account would disgrace her for ever.’

Caleb halted as if his feet were suddenly clamped to the floor. For the first time, he saw the danger with which his impetuous conduct threatened the being he wished to save.

‘What am I to do, then?’ he asked with more humility in his manner than he had yet displayed.

‘What I have told you. Wait as patiently as you can till the morning. Be here at eight o’clock, and I promise to have everything explained to your satisfaction without causing the girl annoyance.’

‘It’s kind of you to think of that, sir.’

‘Show your gratitude by doing what I tell you. Go back the way you came; if you mounted by a ladder, return it to its proper place; and when you come

in the morning bring Culver with you.'

'I can't speak to him about it until I know she is safe.'

'There is no need. You have only got to say he is wanted here. It is better you should take the message than one of the servants. The less gossip we have the more likely the girl is to escape scandal. Good-night.'

'Good-night, and ... thank you kindly.'

The Agitator had never imagined that there would come a day when he should be compelled to speak such words of gratitude to the owner of Ringsford. He obeyed his commands slowly but faithfully: all sense of humiliation was stifled by the knowledge that whatever might be the upshot of the meeting in the morning, the advice given him was sound, and that in adopting it, he was rendering the best service to Pansy.

Mr Hadleigh knew that he had conquered the man, and did not think it necessary even to look at him as he parted the heavy hangings and stepped out on to the balcony. A few minutes after the window had closed, however, he bolted it. That operation had been forgotten by himself during the evening, and had not been performed by the servant, who had instructions on no account to enter the library unless the bell rang.

Resuming his seat at the desk, Mr Hadleigh completed the task in which he had been so unpleasantly interrupted. He sealed the two envelopes, and placed them in the drawer of the safe, which he locked.

'I have done with these things now,' was his mental observation, and yet he lingered over the words, as if reluctant to pledge himself that he should not again look at those records of a sad life. With more firmness he said presently: 'I shall not look at them again.'

He drew a curtain aside and looked out. The moon was shining dimly through a haze; the white space before him looked like ghostland, and it was peopled for him by ghosts of blunders in the past and of hopes disappointed or relinquished as unattainable. If we could live our lives over again! What use?—unless we could start with the bitter experience which inspires the wish. Then how steadily we might steer through the shoals of folly, passion, and falsehood.

In that still ghostland on which he was gazing, there rose a new phantom offering comfort.

‘I will find my happiness in fostering theirs,’ he said, as he turned from the window, and with hands clasped behind him, head bowed, began to walk up and down the room.

Here happened one of those trifles which make and mar existence. He was tripped by a hassock and fell: in falling, his temple struck the corner of the table, and he lay insensible. About the same time there was a strange sound outside like the distant boom of a heavy sea rushing upon a shingly beach: it was the wind forcing its way through the snow-laden trees of the Forest.

Caleb Kersey had reached the village, his hand was on the latch of his lodging, when looking backward, he saw a red glare in the hazy atmosphere. The terrible word ‘Fire!’ rose to his lips, and his landlord—Dr Joy’s successful pupil in the science of economy—heard him. The alarm spread through the village with mysterious rapidity, and whether moved by a desire to render assistance, or mere curiosity and a craving for any unusual excitement which might break the monotony of their lives, groups of men, women, and boys were soon tramping through the snow in the direction of the blaze. The little engine of the village volunteer fire brigade was dragged from its shed, and with a shout the lads started to the rescue.

There had been much rick-burning during the past few months, and it was at first supposed that this was another outrage or accident of the same kind. But the wonder grew, and the pace was quickened when it became known, from messengers who were riding in search of assistance, that it was Ringsford Manor-house which was on fire.

Already there were people at the scene of disaster, and as the broad flames shot out from windows and roofs, there were murmurs of wonder such as one hears at a display of fireworks. The murmurs, however, were those of terror. The luminous glare cast a blood-red shade over the white ground; the snow quickly dissolved, and was trampled into a black puddle by the feet of the gathering crowd.

The fire had got complete possession of the building before it had been discovered. Still, gangs of men were passing buckets of water from the wells, which others heaved on the burning mass; whilst Coutts was vainly exerting himself with an 'extincteur.'

Eager questions passed from mouth to mouth as to the servants and family. All were safe except one—the master of the house, and it was feared he had perished.

Four men bearing a ladder came running from the direction of the stables. The ladder was placed against the portico, which protected it somewhat from the fire. Three of the men drew back from the scorching heat; the fourth, whose form reflected by the light was like that of a giant, dipped a big handkerchief in a bucket of water and fastened it over his mouth. That done he ascended the ladder and reached the balcony. He tried to open the nearest window, but it was fast, and a slight murmur of dismay rose amongst those who watched the rescuer. Placing his shoulder against the casement, he with one vigorous heave burst it open and disappeared.

Suspense lengthened ten minutes into an hour. The man came out carrying another in his arms, and all knew that the other was Mr Hadleigh. The rescuer reached the ladder: instead of descending step by step, he twined his legs round it and slid down, sailor fashion, supporting his burden with the right arm and steadying himself with the left.

It was one of those feats of combined daring, courage, strength, and agility which always win the heart of a crowd, and he was greeted with as hearty a cheer as if he had saved the life of their best friend, instead of one who was at ordinary times no favourite. Nevertheless, there were not wanting expressions of sympathy when the report went round that Mr Hadleigh was dangerously burned, and unconscious.

The young ladies and some of the female domestics had taken refuge in the gardener's cottage, and thither Mr Hadleigh was conveyed, whilst messengers were despatched for Dr Guy and Dr Joy.

By this time the engines from Kingshope and Dunthorpe were at work. The fire had raged within the house for some time before the roof fell; now it came down with a great crash, and the melting snow aided the engines in checking the extension of the flames to the right wing; but it was long

doubtful whether or not that portion could be saved. To this object all efforts were now directed, and there were constant relays of willing hands to work the pumps. By daylight the blackened walls of the main building remained standing, with a smouldering heap inside. Thanks to a thick wall between it and the right wing, the latter had sustained relatively little damage.

The first question asked by the county police was how had the fire arisen. No one could guess, until Coutts Hadleigh said briefly:

‘I believe it was that ruffian, Caleb Kersey.’

He stated his reasons for the surmise, and Caleb was arrested that day on suspicion.

VÆ VICTIS!

WHAT Brennus shouted on the banks of the Allia, and the Romans made into a proverb after him, history has re-echoed ever since in deeds as well as in words. 'Woe to the conquered!' is traced in letters of blood on the sable lining of the golden shield, of which the legend in front, written in lines of light, is 'Hail to the victor!' 'Væ victis!' is the discordant echo of 'Io triumphe!' Woe to the conquered has been the experience of all who have fought either for a principle or a cause; a strip of land to add to the imperial territory, or for the integrity and freedom of the country and for dear life itself. Strike and spare not; kill all, even to the babes and sucklings, the aged men and the young women; tread under foot those who are prostrate; leave to perish by the wayside those who have fallen out from the ranks—væ victis! woe to the conquered, and death to the weak; but hail to the victor, and increase of power to the strong! So goes the world; so has it ever gone in the moral life as well as in the physical; the struggle for existence being as true of thoughts as it is of races.

We must remember the heroic past, when haply times are a little hard to ourselves, and we are bound to suffer in the comparatively mild way of modern days. We have to fight our fight, whatever it may be, as the heroes of old fought theirs, till our cause conquers, or we are convinced that we are on the wrong—not the weaker—side. But till then, we have to endure private loss that humanity may have greater gain; and to remember that conscience is better than victory, and that truth has ever been buffeted before she has been crowned.

The world has this cruel cry for others beside the pioneers and martyrs of a cause. If nothing succeeds like success, so nothing is so fatal as failure; and væ victis! is in truth the sentence recorded against those whose fortunes fail, whose card-houses tumble to pieces, and whose flapping wings of ambition prove themselves to have waxen attachments, which melt in the using and land the poor highflyer in the mud. That fatal settling-day on 'Change: that bad debt made through the bank, and added to indefinitely, on the theory of possible redemption if enough time were allowed and enough rope paid out:

that terrible spell of ill-health which prevented the completion of the book, the painting of the picture, the execution of the order: that failure, that fiasco—and *væ victis* as the commentary!

Væ victis! to the unhappy—to those vanquished by pain and cast down by sorrow. Only a very few compassionate souls care to give their time and strength to the miserable who sit in darkness ever unlifted, and with ashes never shaken off their bowed and melancholy heads. We naturally like the light better than the dark; and perfumed pomade, beneath rosebuds and fine feathered caps, is a more pleasant thing than ashes taken out of the grate and scattered over our hair. We get tired of enduring sorrow. At first, we are keenly sympathetic; but as time goes on, we wonder why the wound does not heal. Our own sympathetic pain has passed long ago; why cannot that of their real hurt? They are always so sad! They take no pleasure in the last fashion, the newest gossip, that good story which is going the round of society, or that smart saying of the ill-natured wit, whose epigram rips the skin from the flesh of his victim as neatly as if it were a rapier. They are always so dull, these poor creatures—it is really impossible to go and see them! *Væ victis!* They are conquered by grief, by loss, by pain; and they must suffer, as all those who are overcome have to learn how.

Væ victis! to the outspoken who cannot back their words by that kind of substantial evidence which passes by the name of legal proof. Thus they are in the power of those against whom they sought to warn the unwary and enlighten the blind. If they cannot so back their words with legal proof, they are conquered, and have to suffer the fate of the conquered—in the law-courts this time, as a change from the battlefield; and with such punishment as belongs to the law of libel to inflict. All the same it was maybe no libel, no falsehood, but the absolute truth that they said; but all the same, too, truth which is only a moral certainty and not a substantive fact to be demonstrated by undeniable evidence, is not to be said without danger, and *væ victis!* to those who cannot substantiate it. By which we are taught the lesson of that silence which is golden; and, in private things, the wisdom of not interfering in the affairs of others. As the Italian proverb has it: ‘A fool knows his own affairs better than a wise man knows those of others;’ and again another: ‘He knows much who, knowing nothing, knows how to keep silent.’ A great deal of trouble is made for ourselves by this interference in the affairs of others. But it is difficult not thus to interfere, when we see all

awry, things which we think a few honest words would put straight. But we must look out for signal discomfiture, unless we can hold the reins we seek to clutch, and make those with whom we have intermeddled see according to our lights and act according to our judgment. If their will is stronger than ours, it is *væ victis!* to us in good sooth!—with the not over-pleasant reflection, that we have pulled that pot of boiling water over ourselves, and in not letting comparative well alone, have stirred it into active ill.

Væ victis! to the unsuccessful aspirant, whether it be for honours, a pass, a post, or for love. No one really pities him. He who droops on the way and falls short in his stride is passed by, and the triumphant reach the goal amid the plaudits of the crowd. Who cares to console a failure? to reconstruct a ruined cloud-tower? to follow after a fading rainbow? *Væ victis!* and let the pale illusions of a dead hope lie where they have fallen! The poor fellow was rejected, was he? Well, he really ought to have been more sure of success before he made the effort! Perhaps he was led on, as you say; but even so, this does not excuse him, nor in any way affect the principle of quasi-certainty before the attempt. When you hazard a great stake, you ought at least to know how to throw the dice, and to be sure that you have counted your cards. It is no use rushing darkly into enterprises for which one has not the material; and to offer one's self for a place, whether of love or honour, without having taken pains to measure one's strength against those forces which oppose, is to be more rash than brave, and more foolish than energetic. To be sure, no one can have who does not try. That is true for itself. But it none the more softens the verdict of the world for him who has failed, nor deadens the echo of the cry raised against him. If we do not try, we cannot get; but woe to those who try—and fail!

Væ victis! also to the stupid, to the poor, to all born conquered by fate even before they have begun to strive with fortune—who are thrust into the battle unlearned and heavily weighted or ever the lists are set. Who pities the dunce? Who cares to realise all the days and hours of hopeless endeavour to get those facts, that task, rooted in the sluggish brain? Is it his fault, poor dunce, that the atoms are slow to move? the nervous fluid insufficient to stir? the blood too thick to run or too poor to animate? He does his best; but as a schoolboy he gets flogged, as well as over-tasked; as a man he gets distanced, and perhaps ruined, unless he has the luck of hereditary bread with butter superadded. But what is not his fault, is nevertheless accounted

to him for blame; and because he is a dunce, he has recorded against him the sentence of disfavour.

What else, too, have the poor?—not the very poor, whose want of bread of their own baking forces the oven-doors of the rich—but the comparatively poor—those who have enough whereby to live, but nothing left for enjoyment—who have the necessities, but not the comforts, nor the graces, nor the pleasures of life? No one pities them, though they suffer in their own way quite as much as those ragged brethren who go cold and hungry for want of clothing and food. These others have to offer a brave front to the world, and to make a little seem a mickle, and not enough a full measure. It is done only at the cost of the night's sleep and the day's peace—at the cost of this thing pinched and that pared, and the deft interlacing of two gaping ends. No one pities all this, because no one realises it; and it may be, as has been known too sorrowfully before now, that those who have almost more than they can do to live decently are blamed for parsimony because they do not live luxuriously.

Perhaps no people deserve more sympathy than those poor gentry whose means fall below the standard of their original condition, and thus fail the present needs and degrade the future position of the family. Gentlefolks born and bred, with the niceness of feeling and delicacy of taste included in that term, they are forced to abandon all the embroideries of class refinement, and to see their children grow up with manners below their own mark, and with an education of less beauty if of as substantial value. They cannot give them the 'advantages' which count for so much. The girls do not learn music from the best masters, and their painting lessons leave much to be desired. The boys have to associate with lads of lower breeding, who teach them rude ways and vulgar expressions, so that all the influence of home goes to undo what that of the school has done. Neither boys nor girls can learn to ride, to hunt, to shoot, like their cousins—the rich branch of this unequally developed family tree; they cannot be taken to the seaside if they fail in health; and, when they are ill, they cannot be nursed so sedulously as if mamma had nothing else to do, and a nurse with brains and experience could undertake the rest. They are heavily handicapped all through, and set to play blindfold with Fortune, who holds loaded dice and marked cards. From the first, *væ victis!* has been recorded against them; and unless they have exceptional power, on which we cannot always count,

they are foredoomed to comparative failure, and to that painful process of winnowing whereby the conquered fall through the meshes into the abyss, and only the victors remain safe on the surface.

So it is: *Væ victis* all through! Nothing succeeds like success, and nothing is so fatal as failure. The law is inexorable, and tears cannot dissolve the links of brass and iron which Fate has forged. The survival of the fittest is only a form of *væ victis*!—the destruction of the weak, and woe to the conquered! But—and herein lies the balm to the sore—it depends greatly on ourselves whether we will be the conqueror or the conquered; whether we will make ourselves strong by endeavour, resolution, self-control, and the cultivation of our reason and common-sense, or let ourselves go to ruin by self-indulgence, weakness of will, unreasonable desires, and the gratification of tastes which we cannot rightfully indulge, and of impulses which, even if good, are unworkable and disastrous. If we resolutely determine that it shall be *Io triumphe!* with us, for the most part we come to our goal; and at the worst we can always mitigate our failure.

ONE WOMAN'S HISTORY.

A NOVELETTE.

BY T. W. SPEIGHT.

CHAPTER I.

‘You are a rank heretic, Mac, neither more nor less,’ remarked the vicar, ‘to say that you don’t care for our lovely Lake scenery.’

‘But listen, my dear friend,’ protested the doctor; ‘I never said anything of the kind. What I did remark was, that your English Lake scenery was not to be compared to our Scotch scenery. It’s pretty—very pretty, but when you have said that, you have said all. If you want grandeur, if you want sublimity, you must go’——

‘To Switzerland and see Mont Blanc,’ broke in Miss Gaisford as she looked up for a moment from her writing.

The doctor shook his head. ‘I have reason to believe that the Swiss scenery has been very much overrated. And then, just consider the expense! I’m told that the innkeepers there are rogues—every man-jack of them. No—no. I’ve been half over the world in my time, and all I can say is, that old Scotia’s mountains and lochs are quite good enough for me.’

The scene was the lawn of the *Palatine Hotel*, overlooking a lovely stretch of Windermere, with the purple-buttressed hills that guard the head of the lake for an imposing background. The time was about four o’clock on a sunny afternoon. Of the three people who had engaged in the conversation just recorded, one was Dr M’Murdo, an army surgeon, the greater part of whose life had been spent abroad. He had just retired from the service on a small fortune left him by a relative, but had not yet quite made up his mind where to settle for the remainder of his days. He and the vicar had been great friends when young men, but had not met for a number of years till to-day, the doctor having arrived at the *Palatine* a few hours ago on a visit to

his friend, who in his turn was spending a portion of his holiday with other friends at the hotel.

Dr M'Murdo was well on towards his fiftieth year. His hair and beard were already grizzled, while his once fair complexion was deeply tanned by many years of torrid sunshine. He was a tall, lean, high-dried man, somewhat formal and old-fashioned in his attire, with an expression of mingled shrewdness and good-humour.

His friend, the Rev. Septimus Gaisford, was about the same age as himself, and had been the vicar of a small country parish in the Midlands for nearly a quarter of a century. He belonged to the homely, unobtrusive type of country parson of which, even in these days of unrest and innovation, specimens happily are still to be found. He looked after the needs of his poorer parishioners both spiritual and temporal, and left the well-to-do pretty much to themselves. He abounded in good works in a quiet unostentatious way, while his Sunday discourses were as homely as himself and such as could always be 'understood of the people.' Like his friend the doctor, he had never ventured on the perilous sea of matrimony.

But the vicar was not without a worthy coadjutor and companion in his parochial labours. His sister, Miss Gaisford, who was ten years younger than himself, not only managed his small household, but looked after such portion of his parish duties as can often be performed better by a woman than a man, while it was even whispered that she occasionally wrote his sermons for him. So that, all things considered, it was no wonder the Rev. Septimus had more than once been heard to remark that 'Pen'—short for Penelope, if you please—was far more useful to him than any curate he had ever had. For the rest, Miss Pen was a bright-eyed, vivacious little body, not in the least inclined to be sanctimonious, but fond of a joke and a laugh, yet with an innate fund of sympathy about her which by some attraction of its own seemed to draw all who were in trouble or difficulty to her side.

On this sunny afternoon, the doctor and the vicar were seated one at each end of a rustic bench in the shade of a leafy elm. The former had his thumb in the pages of a medical review, to which, however, he was paying but scant attention; while the latter was mending his fishing-tackle, for our worthy parson was a genuine brother of the angle. At a small rustic table a little distance away sat Miss Gaisford, busy with her writing, but not so

busy as to preclude her from taking an interest in any topic which the others might introduce.

Presently she looked up, and as if in answer to the doctor's last remark, she said: 'I am quite aware that we poor mortals who have the misfortune to live south of the Tweed are very badly off as regards many things. Still, we do now and then manage to produce an article which even you cool-blooded Northerners can't help admiring, and would find it difficult to excel.'

'The application, my dear madam, the application. To what particular article do you refer?'

'At present I refer to Madame De Vigne, the charming widow to whom you paid such very marked attentions at luncheon.'

'Ah-ha. I noticed something of that myself,' chuckled the vicar.

'Everybody noticed it,' said Miss Pen emphatically.

Dr Mac rubbed the end of his long nose with his review and laughed uneasily. 'Ha, ha! Very good—very good indeed.'

'Come now, Mac, you can't say that you didn't cast sheep's-eyes at her,' put in the vicar blandly.

'Let the pawky Scot deny it if he dare,' said Miss Pen with a shake of her little fat curls.

'Very good, my dear friends; if you choose to make yourselves pleasant at my expense, you are welcome to do so. That I admire Madame De Vigne, I am quite willing to admit. From what little I have seen of her, she seems to me a very agreeable person, and if we could trace back her ancestry, I have no doubt we should find her to be of Scottish extraction.'

'Oh, come, Mac, give us poor Southerners credit for something.'

'Well, I don't mind admitting to you, who are one of my oldest friends, and to Miss Penelope, that I am getting tired of a bachelor's life. I want a home and I want a wife. I have a little money judiciously invested—and—and I thought, in fact—that—that'——

'Don't be bashful, Mac,' chimed in the vicar softly.

‘You thought, in fact, that the charming widow would make you a charming companion for life,’ put in Miss Pen briskly.

‘Perhaps ay, and perhaps no,’ responded the doctor quietly.

‘All I can say is, that you may think yourself a particularly lucky fellow if you succeed in winning her,’ remarked the vicar.

‘Well, well; I know that both of you are friends of Madame De Vigne, and that she and her sister are parishioners of yours. What I should like you to do is to tell me all you know about her, and then leave me to consider what my future course ought to be.’

‘All that we know about Madame De Vigne is very little,’ remarked the vicar.

‘Very little indeed,’ assented his sister.

‘Still, my dear’—to Miss Pen—‘I am not aware that we should be abusing any confidence in telling our friend all that there is to tell, so far as we know it?’

‘There can be no possible harm in that. Besides, it will only be charitable to take pity on the poor man. And now, please not to interrupt me again for ten minutes at the least.’

‘It is now nearly two years,’ began the vicar, ‘since Madame De Vigne and her sister, Miss Loraine, came down to Oaklands, bringing with them a letter of introduction from my London lawyer, a certain Mr Railton, whom I have known for a dozen years or more. How Madame De Vigne came to be known to Railton, or what he knew with regard to her and her antecedents, I had not the curiosity to ask at the time, and I have never since made it my business to inquire. It is sufficient to say that Madame De Vigne had seen advertised as being to let a certain furnished cottage which she thought would suit her requirements; hence her visit to Oaklands. The cottage did suit her. She became its tenant, and there she and her sister have lived ever since, shunning rather than courting such society as our neighbourhood affords, but visiting a good deal among the sick and poor. One day about six months ago, while I was out fishing I encountered a young fellow who was similarly engaged. We met again and again, striking up an acquaintance as brother Piscators have a knack of doing, till finally I invited him to dinner at

the vicarage, on which occasion Penelope there took quite a fancy to the young man.'

'Of course I did,' answered Miss Pen, looking up quickly. 'Any one else placed as I was at the time would have taken a fancy to him. I was just in want of some one to sit for the hero of my next novel, and Archie came in very handy.'

Dr M'Murdo started. 'But, my dear Miss Gaisford, you don't mean to say that you make a practice of introducing portraits of your friends and acquaintances into the stories you write?'

'Don't I though! I shall have your portrait jotted down in my note-book before you are many hours older. I have no doubt it will come in useful one of these days.'

'Good gracious! I hope you won't paint me in very dark colours.'

'Not blacker than you deserve, you may rely upon it.' Then to herself she said: 'Where was I?—Yes—yes,' and so went on with her writing.

'Well, that first visit of young Ridsdale to the vicarage was but the forerunner of several others,' resumed the vicar as equably as though he had not been interrupted. 'It was there that he met Madame De Vigne and her pretty sister, and with the latter he at once fell desperately in love.'

'And the young lady fell desperately in love with him?'

'That is exactly what came to pass. But I'm nearly sure the affair might have been nipped in the bud had not Penelope, with true feminine perversity and reckless disregard of consequences, encouraged the two young nincompoops in their folly.'

'What else could you expect me to do?' asked Miss Pen, without looking up. 'When I see a little romance of real life going on under my very nose, do you think I'm the one to try to put a stop to it? No, indeed. Besides, it supplied me with a lot of hints for love-making scenes; it was what the painters call "a study from the life."'

The vicar shrugged his shoulders, as much as to say: 'You hear the kind of arguments I am compelled to listen to.' Then he again took up the thread of his narrative. 'It was not till after young Ridsdale had become engaged to Clarice Loraine that we discovered he was the son of Sir William Ridsdale,

a wealthy baronet of ancient family. The next thing was to obtain the baronet's consent to the engagement. It would appear that the family estates are not entailed, and, as a consequence, should Master Archie run counter to his father's wishes, the latter can dispose of his property in any way he may think best. Well, the all-important missive was written and posted to Mentone, where the baronet was at that time residing. The answer was—well, what do you think it was?’

‘A peremptory order by the first post to the young man to break off the engagement.’

‘Nothing of the kind; but a visit one afternoon at the vicarage from a certain Colonel Woodruffe, who had come as plenipotentiary from the baronet. Sir William was an invalid, and could not travel, so he had deputed the colonel to act in his stead. The father had no doubt in his own mind that his son had fallen into the toils of an adventuress, and the colonel's instructions were to break off the engagement at any cost, and take Master Archie back with him.’

‘A sensible man that Sir William. And how did the affair end?’

‘In a way that you would hardly guess. The gallant colonel, instead of carrying out his instructions, and breaking off the engagement between the young people, ended by falling in love himself with the fascinating widow and proposing marriage.’

‘A change of front with a vengeance! And the answer?’

‘A rejection.’

‘Check for the colonel.’

‘But, old bachelor though you are, Mac, I daresay you know quite enough of the sex to be aware that a woman's No is not always final. At anyrate, the colonel, who is really a very fine fellow, is evidently a believer in that doctrine, seeing that five days ago on his way to Scotland he stopped here for an hour, had an interview with Madame De Vigne and renewed his offer.’

‘And the answer to his second offer?’ queried the doctor eagerly.

The vicar shook his head. ‘Pen, perhaps, can tell you more about that than I.’

Miss Pen looked up quickly. 'The answer is to be given him to-day.'

'To-day!'

'The colonel will call here this afternoon on his way back from Scotland, when Madame De Vigne has promised that he shall have her final decision—Yes or No.'

'So that, my dear Mac,' said the vicar with a smile, 'all things considered, your chance of winning the widow does not seem a very promising one.'

'Well, well,' answered the doctor sturdily.

'If a better man than Sandy M'Murdo wins the fair prize, why then I'll—I'll be his best-man at the wedding.'

For a minute or two nothing was heard save the busy scratching of Miss Gaisford's pen.

'How will this do, Septimus?' she asked presently, and with that she began to read from her manuscript.

"Her eyes of tenderest April blue glance up shyly into his dark volcanic orbs, in which there is a half-smothered fire that causes her heart to flutter like an imprisoned bird. A moment later, and her slender willowy form is swept up in a passionate embrace by those stalwart arms, and Love's first burning kisses are showered on the sweet rosebud of her lips."

'Rather tropical, is it not, my dear?' hinted the vicar mildly.

'Oh, there's nothing namby-pamby about my readers, I assure you,' answered Miss Pen with a merry laugh. 'They like their love-making warm—and plenty of it.'

For ten minutes longer the busy scratching went on; then Miss Gaisford laid down her pen with a sigh of relief. 'There—not another line to-day,' she said. 'Now that I have got my hero and my villain in the midst of a terrific encounter on the verge of a precipice, I can leave them there for a few hours in comfort.'

'That seems rather cruel to the pair of them,' remarked Dr Mac.

'Oh, heroes and villains are used to that sort of treatment.—But I hope you will keep my little secret a secret still, doctor. If it were to reach the ears of

any of the goody-goody people at home, that the parson's sister writes foolish love-stories for young people, what hands would be uplifted in holy horror—what ejaculations over her backslidings would be whispered across half the tea-tables in the parish! Neither the squire's wife nor Lady Pinchbeck would ever speak to me again, and what, oh! what would existence be worth under such terrible circumstances!'

'My dear madam, you may rest assured that your secret is perfectly safe with me.'

'It will be a bad day for the poor of my parish when Penelope gives up writing her love-stories,' remarked the vicar, who was busy with his tackle book. 'Every penny she earns goes to buy blankets, and coals, and such-like comforts for those who have no money to buy them for themselves.'

'My dear Septimus!' exclaimed Miss Gaisford with a flaming face.

'My dear Pen!—Now that Mac has been taken into our confidence as regards one side of the question, it is only right that he should be made acquainted with the other.—But here come our two truants,' added the vicar a moment later, as Mr Archie Ridsdale and Miss Clarice Loraine, looking somewhat conscious, emerged from one of the winding walks, and came towards the hotel, each of them laden with a quantity of wild-flowers, ferns and grasses.

'The lovers, eh,' said Dr Mac, half to himself. 'A very bonnie young lady—very bonnie indeed.'

'We were just thinking of sending the bellman round,' said Miss Pen, as the truants came up. 'Ting-ting-a-ling. Lost since early this morning, a pair of sweethearts. When last seen, he had his arm round the waist of she, and she had her head on the shoulder of he. Whoever will'——

'Stop, do!' cried Miss Loraine, as she dropped her ferns and grasses on the table and stuffed her fingers into her ears.

'We have been botanising,' observed Mr Ridsdale with the most innocent air imaginable.

'And a pretty lot of rubbish you seem to have brought back,' remarked Miss Pen.

‘Rubbish, indeed! And not one among them without a long and beautiful Latin name of its own.—Ask Archie.’

The vicar rose, and addressing the doctor, said: ‘Allow me the pleasure of introducing you to Miss Loraine.—Clarice, my dear, this is Dr M‘Murdo, a very old friend of mine.’

‘I had the pleasure of being introduced to your sister this morning, Miss Loraine, and now the pleasure is doubled,’ said the doctor with a touch of old-fashioned gallantry.

‘I am happy to make the acquaintance of any friend of Mr Gaisford,’ answered Clarice with a smile and a little blush.

‘Mr Archibald Ridsdale—Dr M‘Murdo,’ said the vicar. The two men bowed. ‘A capital fellow to know, so long as you are in good health, Archie, but a fellow to fight shy of the moment you begin to feel yourself out of sorts.’

Suddenly a shrill whistle was heard. ‘Here comes the steamer,’ exclaimed Miss Penelope. ‘I’m going by it as far as Bowness. Any one going my way?’

‘I will walk with you as far as the landing-stage, and see you safely on board,’ answered her brother.

‘That way will suit me as well as another,’ added Dr Mac.

As the two men turned to go, Miss Pen drew Clarice aside. ‘Any news?’ she whispered.

‘None,’ whispered back the girl with a doleful shake of the head. ‘If Archie and I only knew the best or the worst! It is this suspense that’s so hard to bear.’

‘It seems hard to bear at present, but it will be delightful to look back upon, by-and-by.’

‘O Miss Penelope! how can that be?’

‘Just now you are in the middle of the first volume of the romance of your life. Now, I should like to know how a romance can be worth anything without suspense, without mystery, without your being unable to guess what may happen from one hour to another.’

‘Penelope, you will certainly miss the boat,’ called out her brother, who was already some distance away.

‘We will talk more of this anon, my dear,’ said Miss Pen hurriedly. ‘Meanwhile, don’t get downhearted; all will come right at the end of the third volume: it always does.’ And with a nod and a smile, the bright little woman tripped off after her brother and the doctor, and presently the trio were lost to view down the winding path that led to the landing-stage and the lake.

POPULAR AMUSEMENTS IN GERMANY.

WHEN one thinks of the downright substantial way in which English people amuse themselves, with cricket, football, lawn-tennis, rowing, foot-racing, leaping, &c., all over the kingdom, it is strange to find so little of active sport of the kind among our muscular cousins of the Vaterland. To be sure there are boating clubs and athletic societies, and gymnasia in the schools and elsewhere; and soldiers are exercised in gymnastics, but there is too much of military stiffness about it. Amusements of a passive character find most favour. To 'make' a walk in a leisurely fashion, to drive in an open conveyance, or to sit down in a shady grove to listen to a band playing, is the most acceptable mode of enjoying relaxation. With beer at a penny-farthing a glass, and a tolerable cigar for a halfpenny, and a military or string band discoursing sweet music, the time passes pleasantly enough. It makes a great difference when you have not to pay very dearly for your whistle, and they certainly know how to get the most for their money in the land of the Teuton. Many houses of refreshment, even in towns, have gardens or courtyards thickly planted with trees, so that their branches meet overhead and form a pleasant and inviting shade. A large shed, too, is provided, open on the garden-side, in case of rain. Frequently, music is introduced, and on these occasions, an extra halfpenny is charged on the beer, to cover the expense of the entertainment. On Sundays and festivals, there is music, beer, wine, and tobacco everywhere. And yet these people know how to amuse themselves without going to excess. Sometimes a tipsy man is seen, but rarely till very late at night, and the occurrence is so infrequent, that, compared with the usages of our own country, it is quite remarkable.

We have more than once asked the question: 'Is it that the beer is weaker, or that the German heads are stronger than the English?' and we have been told: 'Perhaps it is a little of both.' Perhaps, too, there is something in the fact that there is in some respects less class distinction in Germany, and the middle classes may be seen sipping their wine or coffee in the same place with their hard-working brethren. Perhaps the national and natural good opinion, self-respect, or self-esteem—call it what you will—of the German

helps to keep him straight; and then he takes his creature-comforts in a staid, stolid, philosophic way. Noisy fellows there are, of course; but they do not squabble and fight, as a rule; the utmost they are guilty of being the national practice, even at midnight, of singing rollicking choruses, to the great disturbance of peace-loving, law-abiding, slumbering citizens. The fact that soldiers are permitted to wear their side-arms constantly, speaks volumes for the sobriety of the men as a class, and redounds to their credit.

A *Turnfest* or athletic festival, generally held on a Sunday, is a great affair, often the event of the year in a small provincial town. There is a wonderful display of flags everywhere; and in the afternoon, a procession of the competitors and visitors with bands and banners and every variety of costume, the medals, badges, and ribbons of former contests being worn with great ostentation. The most is made of everything; and shouting and singing and cheering, and dust and noise, seem to be the order of the day.

Rowing-matches provoke immense enthusiasm, and a regatta is an affair that induces the keenest interest. A people with so much love for the wonderful and so much regard for themselves, cannot help throwing into such occasions an amount of enthusiasm and national pride as would do credit to the Oxford and Cambridge boatrace itself. The members of the Boat Club are the heroes of the hour, and their costumes the object of great admiration. Though not so peculiar as the French in this respect, our German neighbours are nevertheless great in their 'get-up' for every particular sort of occupation or sport that they engage in. If a man brings down only one snipe in a day, he looks tremendously cut out for business notwithstanding, and appears every inch of him a sportsman when going to or returning from the 'hunt,' as he calls it.

The exercise of riding cannot be properly accomplished without a complete and appropriate rig-out, so that even when he is not actually on his horse, the equestrian gives to all the world the assurance of a man at home in the saddle. If spurs and jack-boots do not make a rider, they at anyrate look very much in earnest. Never did a be-uniformed people more thoroughly believe in the dignity of dress and the necessity for effect than the Germans.

However we may smile at the eccentricities and oddities of the Germans, we must admit that they beat us in the provision of cheap music for the people, most of whom understand and appreciate it. Every school-teacher is

bound to be a musician and to pass in music, so that the people have a chance of learning from childhood.

If the Prince of Wales succeeds as well in popularising the study of music as his father did in popularising art, we may hope to see before long a great reformation in the morals of our own people; and the wandering German bands, composed of the worst players in their own country—where they would not presume to play in public—will no longer be tolerated in England, because the taste of the people will be educated above such wretched performances. Good music, then, everywhere is what is wanted to enable the lower classes to enjoy themselves rationally, and no better means of promoting the sobriety of a nation can be devised. The more the masses are leavened with a knowledge and love of music, the more indeed we imitate the Germans in this respect, the less necessity there will be for restrictive measures in the way of ‘local option,’ and the lighter and easier will be the work of temperance reformers. A great reform will have been effected. If music can charm savages and snakes, it can do much more for our toiling, amusement-lacking countrymen and countrywomen.

The theatre is much patronised in Germany, the prices being cheap, the music good, and the performances fair. The play begins at seven, and ends about nine or half-past. Concerts, too, both instrumental and vocal, are frequent; every town has its Choral Society, and every district its Choral Union, so that there is never any lack of vocalists of both sexes for the performance of an oratorio or the celebration of any great occasion.

Where a people is satisfied with simple pleasures, these can, of course, be provided at little cost. Children have their swings, climbing-poles, bowling alley, merry-go-rounds, horizontal bars, &c., in the public gardens; and when one sees groups of officers deeply interested in the game of dominoes, it does not cause so much surprise to witness a huge whirligig worked by horses or steam, where servant-maids and soldiers are driven round and round upon painted wooden horses to the enlivening strains of a barrel organ, aided by a cornet or two and the universal drum. This sort of affair is a great attraction to the masses, and being generally placed near a beerhouse, admiring friends sit round on benches with their beakers of frothy beer, their cigars or large pipes, waiting to take a turn on the machine after a time. Music everywhere seems to be the rule—there can never be too much of it to please the people. Wherever there is a company of soldiers

marching along the road without a band at their head, they make up for the deficiency by singing popular melodies and martial songs, keeping time with their feet; and this always gives strangers a favourable impression of the hearty, happy, and even merry German soldier.

A torchlight procession headed by a band of music is a favourite mode of making a demonstration on any particular festive occasion; and last, but not least, is the highly popular serenade. The Choral Societies of a town will unite—as was the case in Darmstadt on the night before the marriage of the granddaughter of Queen Victoria, the Princess Victoria of Hesse—to the number of three hundred men, and parade before the residence of the person they delight to honour, each member carrying a lighted Chinese lantern at the end of a stick. A selection of popular songs and glees suitable to the occasion is sung, and in many instances the melodies and words are peculiarly fitting. One might write a whole chapter about the amusements of the Carnival time, the masked balls and street displays; and although these affairs are somewhat stolid and quite decorous in character, they contain the elements of simple fun, innocent recreation, and hearty enjoyment. Here, again, music is an important factor, for it enters into everything, and forms the beginning and the ending of every variety of popular amusement.

PRINTERS' ERRORS.

IT must be, to say the least of it, annoying to the speaker or writer possessed of any degree of sensitiveness, when he finds his plainest statements, or it may be his most carefully prepared flights of fancy, turned into nonsense by the substitution or omission of a letter in the printing; and by some unhappy chance it often seems that the mistake is made in just such a manner and place as will do the most mischief. The unlucky poet who wrote,

See the pale martyr in his sheet of fire!

must have been completely crushed when the line appeared as—

See the pale martyr in his shirt of fire!

We can sympathise also with the poet who, writing of his love, asserted that he had 'kissed her under the silent stars,' and found the compositor made him state that he 'kicked her under the cellar stairs.' True, it has been doubted if these two poets ever existed; but others, of less mythologic fame, have suffered as badly at the hands of the printer. Burns, in a cheap edition of his works, is made to say,

Oh, gin my love were yon red *nose*.

A well-known temperance lecturer was indignant at finding the sentiment ascribed to him that 'drunkenness was jolly,' whereas he had declared that it was 'folly.'

For the explanation of many of these blunders it is necessary to bear in mind that in setting up the type the compositor has the various letters arranged in separate divisions of his case and selects them one by one as required. Habit enables him to do this with extreme speed and accuracy; but it will easily be seen that the presence of a wrong letter in a division or a dip into the wrong box may occasion one of these unhappy blemishes. In this manner we find oats rendered 'cats;' songs, 'tongs;' poets, 'posts;' or as once happened in the report of a railway accident, 'confusions of the limbs'

for ‘contusions of the limbs.’ And by the substitution of *n* for *h*, a newspaper report was made to state that ‘the people rent the air with their ten thousand *snouts*.’

The blame, however, does not always rest with the compositor. Incorrect spelling and slovenly writing have much to answer for, especially in the case of proper names and quotations from foreign languages. Boerhaave becomes ‘Boershawe;’ and *Et tu, Brute!* ‘Eh, the Brute!’ Authors should remember that the proof-reader is fallible; he is not, as is sometimes expected, a ‘Universal Compendium’ of facts, people, places. If a passage reads clearly and grammatically, although conveying anything but the sense intended, it is not to be wondered at that the error is often undetected until too late. Much surprise was occasioned by Sir Archibald Alison, in his *History of Europe*, including amongst the persons present at the funeral of the Duke of Wellington the name of ‘Sir Peregrine Pickle.’ There can be little doubt that the author had made an unconscious slip, intending to name Sir Peregrine Acton. Sir Thomas Brassey having referred in a speech to the *Golden Treasury of Songs and Lyrics*, the compositor transformed the title into the ‘Golden Treasury of Soups and Cynics.’ A report in a Manchester paper of a recent dramatic performance mentioned the well-known farce of *No. 1 Round the Corner* under the amusing and suggestive title of ‘*No One Round the Corner*.’

Mistakes in punctuation, such as the omission or misplacing of a comma, may cause serious alteration to the sense of a passage. The contract made for lighting the town of Liverpool by wick lamps, during the year 1819, was rendered void by the misplacing of a comma in the advertisements, thus: ‘The lamps at present are about four thousand and fifty, and have in general two spouts each, composed of not less than twenty threads of cotton.’ The contractors would have proceeded to furnish each lamp with the said twenty threads; but this being only half the usual quantity, the Commissioners discovered that the difference arose from the comma following, instead of preceding, the word ‘each.’ In the following instance, it was no doubt a bachelor-compositor who, in setting up the toast, ‘Woman, without her, man would be a savage!’ got the comma in the wrong place, and made the sentence read, ‘Woman, without her man, would be a savage!’

All the above-mentioned errors may fairly be ascribed to carelessness and mischance. Others, however, are on record which have been committed

knowingly and intentionally, and so can scarcely be classed as errors. They have been mostly connected with Biblical matters, and intended to further party interests. It is said that Field, a printer of the time of Charles I., was paid fifteen hundred pounds by the Independents to alter a single letter in the third verse of Acts VI., so as to make the word we read 'ye,' and so give the right of appointing pastors to the people, and not to the apostles. The deplorable state of the press in Field's time may be realised from the fact that Bishop Usher, on his way to preach at Paul's Cross, asked at a stationer's for a copy of the Bible; and on examining it, found, to his astonishment, that the text from which he was about to preach was not in the book! The well-known 'Vinegar Bible' was published in 1717, and obtains its name from the Parable of the Vineyard being printed as the Parable of the *Vinegar*. One of the most wilful alterations of the text, and one which cost its perpetrator her life, was committed by the widow of a German printer. One night, while an edition of the Bible was being printed in her house, she took the opportunity of altering the word *Herr* into 'Narr,' making the verse read, 'he shall be thy *fool*,' instead of 'he shall be thy *lord*.'

The celebrated Bibles of Sixtus V. are eagerly sought for by collectors. Their sole fame is the multitude of errata which crowd their pages, notwithstanding that His Holiness Sixtus V. carefully superintended every sheet as it passed through the press, and finally prefixed to the first edition a bull forbidding any alteration in the text.

A curious jumble appeared in a cabled critique of Mr Irving's acting on one of his appearances in the States. Instead of saying that 'the taste for Irving, like that for olives, must be cultivated,' the critic was represented as giving utterance to the incomprehensible assertion that 'the toast for Irving, like the toast for olives, must be cut elevated.'

A Glasgow divine, and one of Her Majesty's chaplains in Scotland, was lately reported as saying that 'personally he violated the Lord's Day as much as any member of the Court.' 'Venerated' was probably the word actually employed by the reverend gentleman.

Similarly, in an edition of *Men of the Time* published in 1856, the then Bishop of Oxford is thus described: 'Oxford, Bishop of (Rt. Rev Samuel). A more kind-hearted and truly benevolent man does not exist. A sceptic as

regards religious revelations, he is yet an out-and-out believer in spirit manifestations.' This description really belonged to 'Owen (Robert) of Lanark.' The edition was soon suppressed, but not before the Bishop had possessed himself of a copy for his private library.

Although both the chaplain and the bishop had reason to complain of their treatment, it must have been considerably more astonishing and mortifying for Herr Franz Liszt, who is still delighting his musical admirers with his productions, to find that in Haydn's *Dictionary of Dates* (1870) he is represented as dying in October 1868.

The Queen's Speech at the opening of a recent session of parliament was hurriedly published in a Scotch newspaper without being revised by the press-corrector, and Her Majesty, instead of saying that certain negotiations 'will, I doubt not, lead to satisfactory results,' was reported as saying, 'will, I doubt, not lead to satisfactory results.' So much for the misplacing of a comma!

With increased literature have come better systems of 'reading' and correcting, and greater accuracy has been attained. Such mistakes as above quoted are exceptional, and the morning newspaper may now be read week after week with but few misprints. A good example of accuracy in printing tables of figures—in which it is so difficult to avoid errors—may be seen in *Bradshaw's Railway Guide*, every monthly part of which contains from seven to eight hundred thousand figures. Astronomical and mathematical tables require great care in printing, and very few are issued which can be relied upon as absolutely correct. Charles Babbage superintended the production of a set of trigonometrical tables in 1827 which perhaps stand unrivalled in this respect. They were prepared for the Ordnance Survey of England and Ireland, and a limited number of copies printed—it is said only thirty. They consist of tables of logarithms and log. sines, tangents, cosines and cotangents, *to every second*. Roughly speaking, they contain about six millions of figures. The proof-sheets were revised by several sets of readers, and were carefully examined and compared with other tables no less than seven times. From Mr Babbage's preface, we learn that after the final stereotyping, seven errors were found in the logarithms and one in the differences. These being corrected, the stereotyped sheets were hung up in the Hall at Cambridge University, and a reward offered to any one who could find an inaccuracy. Since their first issue in 1827, no error has been

discovered, and it may reasonably be concluded that they are absolutely correct.

‘THE KING COUNTRY.’

THE recent visit of King Tawhiao to this country has awakened a deep and widespread interest in New Zealand and its inhabitants. The Maori king has been fêted and feasted far beyond his desires, and has paid the penalty of greatness. But he has suffered martyrdom in a good cause; and if he has not succeeded in bringing the wrongs of the Maoris home to the Pakehas, he has certainly aroused a very general curiosity as to the character and resources of the mysterious ‘King Country’ of which he is the titular chieftain. The publication of Mr J. H. Kerry-Nicholls’s volume of travels—entitled *The King Country, or Explorations in New Zealand* (London: Sampson Low)—in this unknown region is, therefore, peculiarly well timed. Until within the last few years, ‘the King Country’—as a vast tract of the finest land in North Island is still known—was a *terra incognita* to Europeans. Rivalries of race constituted a barrier more impassable than the *aukati* line itself, which separates the Maori lands from the European portion of the colony, and is marked on the one side by the farms and homesteads of the settlers, and on the other by the huts of the natives. In its vast forests, over its precipitous mountains, along its trackless plains, the natives alone wandered. It was an *imperium in imperio*, a fastness in which aboriginal sovereignty sat enthroned, deaf to all the offers of civilisation.

It was through this unknown country that Mr Kerry-Nicholls pushed his way in company only with an interpreter. With but three horses, which were ultimately reduced to two, he accomplished more than six hundred miles of travel, discovered many new streams, penetrated almost inaccessible regions of mountainous forest, found extensive plains, traced the sources of three of the principal rivers of the colony, examined the unknown shores of its largest lake, ascended one of its highest mountains, experienced degrees of temperature varying from eighty degrees in the shade to twelve degrees below freezing-point, and successfully traversed from south to north a territory with an area of ten thousand square miles, which had been, from the early history of the colony, rigorously closed to Europeans. It is scarcely necessary to add that the records of these wanderings constitute a singularly interesting volume of discovery and adventure, which can hardly fail to

prove of some practical utility to the colony, and a welcome contribution to geographical science.

‘The King Country’ comprises one of the finest tracts of land not only in New Zealand, but in some respects in the southern hemisphere. It is much more than merely picturesque, and in this respect alone it can compete with the finest scenery the world can produce. Its natural advantages are of the highest order. It is throughout well watered, and in parts exuberantly fertile; while even the mountains are richly clothed with forest trees and shrubs. The southern portion is drained by the Whanganui River, which is fed by many tributaries, flowing from the highest mountain-ranges in the central and southern divisions of the island; while the Mokau River and its affluents flow from the central region to the coast. In the north, the Waipa and numberless minor streams flow from the mountains into the Waikoto River itself, which has from time immemorial been renowned in Maori fable and romance. Again, more than a dozen streams flow into Lake Taupo, an immense reservoir, some twenty-four miles long by fourteen broad, which lies almost in the midst of the central tableland; while the Waikoto is the only effluent river. During the rainy season, the waters of this lake, having only this one outlet, rise rapidly; and with the continuance of heavy winds, its waves are lashed into fury, and break upon its shores with all the force of a raging sea.

Geologically speaking, this district presents problems of surpassing interest. Here can be seen side by side the relics of the stupendous action of volcanic fires, and of the scarcely less potent force of the glacier. It is contended that the formation of North Island must be attributed to submarine volcanic eruptions, which, perhaps by slow degrees, perhaps rapidly, forced upwards the Taupo tableland. These fires bursting again through the plains, caused mountains to rise up in the form of serrated ridges and truncated cones, which poured out streams of lava and enormous deposits of pumice over the surrounding country. Probably the basin of the lake was once an active crater, from which the first vast pumice-plains flowed; while later, the mighty Ruapehu, and, when it became extinct, the still active Tongariro, became the outlet of the volcanic fire. That this element is still largely active in many parts of the country is shown by the geysers, solfataras, fumaroles, and hot springs which form one of the characteristic features of North Island. Thus the ‘lake country’—as the district round Lake Rotorua is

commonly known—is ‘a region of eternal fire.’ The conditions of existence here are certainly novel. The natives bathe in such of the thermal springs as are of suitable temperature, at all times of the day, and in a very primitive fashion. In others, again, they cook their food and warm their houses by the same means.

These natural phenomena occur in many other districts, such, for instance, as that round Lake Rotokawa, and the districts near the native settlement of Tokanu, on the south side of Lake Taupo; while on the northern slope of Tongariro are some of the largest and most active boiling springs in the country. Moreover, the mountains possess all the rock formations in which gold, coal, iron, and other minerals are found to exist. Thus, the Kaimanawa Mountains, which are situated in almost the centre of the island, and stretch across the great central tableland to an extent of eighty miles, offer a peculiarly rich field to the geologist. Mr Kerry-Nicholls reports the existence of abundant auriferous indications, and confidently expresses his opinion that here lies a probable Eldorado.

It is not our purpose to do more than mention the adventurous journeyings recorded in this volume. Frustrated, owing to disputes between the natives and the colonial government, in an attempt to enter ‘the King Country’ from the north, the traveller successfully essayed to pass the southern boundary-line. By throwing himself upon the good-will of the natives, going fearlessly among them, respecting their customs, and following as nearly as possible their mode of life, and, in fact, for the time becoming a Maori, he succeeded at a time when many Europeans would not only have failed, but probably have paid the extreme penalty for their rashness. In one case, it is true, he only achieved his purpose by dint of exercising the greatest secrecy. Tongariro is *tapu*, or strictly sacred, in the eyes of the Maoris, and could only be ascended by a European without their knowledge. This is one of the most perfect volcanic cones in the world. But the resources of this rich district, from whatever point of view they may be considered, are still awaiting development. Its flora and fauna have still to be collected and classified. Its agricultural and industrial resources are still unknown; but we have evidence to show that these are worthy of attention.

THE MOULMEIN ELEPHANTS.

SOME time ago there was a discussion in the learned journals regarding 'intellect in brutes;' and I thought then, as I think now, that much of the controversy depended on the definition we assign to the word 'intellect.' Some say that it is merely an exaltation of the natural instinct of the brute; others, that it is an exhibition of true reason. But then, what is instinct? Some arguers mystify their hearers, and exhibit their own ignorance of the subject by replying: 'Instinct is only that in animals which we call reason in man.' Well, this is not the place to argue the subject; but I shall exhibit certain facts, observed by myself, in the behaviour of the elephants employed in the Moulmein timber-yards, and leave my readers to judge whether they were due to instinct or reason.

Anchored abreast of Aga Synd Abdul Hosein's timber-yard, and within bare swinging distance of the shore, I had ample opportunities of minutely observing and recording the marvellous illustrations of the elephant's intelligence. These animals are largely employed in the timber-yards, and their functions consist in helping to embark and disembark the huge teak-logs, or move them about the yard; in fact, without them work would be at a stand-still. What struck me at once was the wonderful combination of enormous power with the gentlest, most loving docility. Here were huge logs being moved about as if they were matches, and yet with the utmost regard to any one in the way. A case in point. We were landing one day at the Aga's wharf, and found that the ebbing tide had left a thick layer of treacherous slime on the wooden slope, rendering it impossible for a lady to land. Seeing this, the manager called out to a mahout or driver, and in a moment his elephant pushed a log down the slope, just stopping short of the boat, and affording the lady a dry surface to step out upon. There must have been intellect in this act; for the great log was not pushed down at random on the wharf, nor into collision with the boat, but exactly at the right spot and into the right place.

All elephant-work is performed either by the trunk or right foot in pushing; by trunk and tusks combined, as in carrying logs; or by the strength of the

whole body in dragging. Dragging-elephants are furnished with a light wooden pack-saddle, on which the mahout sits sideways, and to which the traction-chain is attached. This is Y-shaped, the leg being greatly prolonged and ending in a hook. Let us watch the handling of this log—twenty feet long, by sixteen inches square—which has to be dragged across the yard. The chain is passed round it by an assistant and then firmly hooked; and now the elephant has to do the rest. His first action is to get his hind-legs well within the V of the chain, and then he starts, the log helplessly following. Arrived at its destination, the elephant disencumbers himself of the log by unhooking the chain with the finger of his trunk, and then pulling it from under the log, or pushing the latter to one side. Now, mark what was involved in this apparently simple operation, the sole guidance to which was either the voice of the mahout, a pat from his stick, or a tap from his heels. First, there was the getting inside the V. Why did the old fellow do that? He has learned from experience that if he did not, traction would be interfered with, and his legs rubbed by the chain. Was not his action, therefore, dictated by reason? Secondly, there was the unhooking of the chain, which instinct never could have prompted. Imitation was at work; the elephant had seen that the unhooking of the chain liberated the log, and had learned to follow the example; showing thereby the domination of reason.

Here are some logs being adjusted on the wharf-slope. Note the ease with which the elephant pushes each into its place with, apparently, the slightest movement of trunk or foot. Mark this one, which, by a greater than necessary exercise of force, has become tilted up against its fellow. The elephant has noted it too, and half-kneeling, and getting his tusks under it, he pulls the log backwards a little, and it drops square with its fellow. What dictated that action? Mathematical order and precision belong to the bee, and are said to be instinctively implanted in that humble animal; but could there have been anything in this elephant's antecedents to have prepared it for rearranging a dislocated log? Surely the impulse seized it at the moment, and must have been due to a sense of order or tidiness implying the presence of reason. Odd lengths of log, varying from four to six feet, are carried about the yard by elephants, a species of work which is distasteful to them, as exhibiting their awkward points. Indeed, they evidently feel degraded by it, for they set to work with an air of resignation quite foreign to their shifting or dragging feats. There they rejoice in their great strength, and are fond of exhibiting it. Here, little strength is needed; but the

operations involve roughish treatment of the nose, and we know that all animals, including man, are very particular as to how their noses are handled.

The elephant is proud of his strength, but sensitive with regard to his trunk, especially when that delicate organ is brought to bear upon any rough work; and as the securing of a log between the trunk and tusks necessitates a large amount of awkward movement, I noticed that the selected elephant approached the job with reluctance. Resigned to his fate, he half-kneels before the odious object, and gets the points of his tusks under it; then he wriggles it up the ivory tramway with his trunk, and secures it therewith *in situ*. His troubles now commence; for on rising to his feet, the hateful log, obeying the laws of gravity, at once tends to drop; and to obviate this, the poor brute has to raise his head on high. Thus constrained, he commences his march with slow stateliness, as if to make the best of a bad job, and not let the world at large know that he is virtually walking blindfold. And so he is; for the elevation of his head upsets the axis of his vision, and he has to walk more by touch than by sight.

I have thus attempted to describe the main duties which elephants have to discharge in the timber-yards, and I have mentioned that they are guided by the voice, stick, or heels of the mahout. Watching them from shipboard, you are quite close enough to note every movement of the animal, but not sufficiently near to catch the signal, so that the elephantine actions seem purely automatic, and therefore the more astonishing. But when you are alongside the animal, and can see and hear the simple signals under which he works, you are equally astonished at the thorough manner in which he understands what he is expected to do, and the very little prompting he requires.

On one occasion, I arranged with a mahout to bring up his elephant to where I was standing, that I might indicate the work to be done, the mahout to be absolutely silent. Standing by a six-foot log, I beckoned to the mahout, and up came the elephant. Arrived at the spot, and being without chains, he must have opined that dragging was not intended. There remained, then, pushing or carrying, the latter operation being the one which the sagacious creature saw was intended, for he proceeded at once with his awkward preparations for carrying it away. Throughout this test the mahout was absolutely silent, and, as far as I could see, quite passive. The

result of it was that the elephant divined: what I, a stranger, wished it to do, and did it.

On another occasion, I applied the test to a difficult object, an eighteen-inch cube of teak, which the dear old fellow at once arranged to carry off; but how to do it, he could not at first determine. As his tusks diverged more than eighteen inches, they were no support, and the many sharp corners of the cube sorely tried the delicate trunk. After some failures, he managed to seize the fragment by the centre, and then raise it up below the tusks against his lower lip. As he had virtually accomplished the task, I discontinued the experiment, expressing my satisfaction and delight to the manager, who somewhat damped my ardour by informing me that the mahout, while abstaining from use of voice or stick, might have conveyed his wishes to the elephant by pressure with his heels!

But a moment's reflection increased my admiration at the elephant's intelligence, for, allowing that the mahout's heels *had* pressed his side, how could such pressure inform him that he was neither to drag nor push, but carry? Surely the mahout could not have possessed a code of pressure-signals with which he had indoctrinated the elephant in prospect of curious visitors. If he had, then it must have included voice and stick signalling as well, to either of which I might have resorted. No; I believe that the elephant acted independently of signals, and reasoned on what he had to do, by what was laid before him.

Hitherto, we have seen the elephant in the yard; let us follow him into the mill, and there admire the triumph of reason over instinct. We all know how naturally timid and nervous the elephant is, and how susceptible to noise. Well, watch this noble old fellow solemnly dragging in a huge log to the sawmill. Onward he moves, undismayed by the horrid panting of the engine or the screeching of the saws. Instinct would have tempted him to turn tail and flee from the noisy turmoil; reason keeps him at his task, confident that amid the uproar and apparent confusion, perfect order and safety prevail. And so, with flapping ears and swaying trunk, he yields up his log to the grip of the remorseless saw, and goes off unconcerned to find a fresh victim. It was very pleasing to see that the Aga's elephants were kindly treated and well cared for; the goading *ánkūs* (iron hook; and prod) was nowhere to be seen. A daily as well as a rigidly observed weekly rest was secured to them; besides, their cleanliness is well looked after; and morning

and evening they are taken into the river to be well scrubbed, the termination of each bath being a triple dive, which they enjoy immensely.

On a second visit to Moulmein, I noticed another totally different illustration of confiding reason in the elephant. We were at anchor in the river in a strongly flowing tide, when a deeply sunk raft laden with green forage, two men, and two elephants, swept swiftly past. The elephants stood motionless and quite unconcerned, knowing that they were under secure pilotage, and quite safe as long as they remained quiet. If they had proved restless, the raft must have come to grief.

GUM-ARABIC AND THE SOUDAN.

According to the *Scientific American*, the gum-arabic supply appears to have been in a great measure cut off owing to the state of affairs in the Soudan. It says: 'Gum-arabic comes almost exclusively from the Soudan, and owing to the operations of El Madhi, there have been no receipts of any consequence for a year past. In confectionery it makes about thirty per cent. of the best quality of gum-drops, marsh-mallow, and jujube paste. The annual supply from the Soudan has heretofore been from twenty to twenty-five thousand bags of four to six hundred pounds each, and there is usually a stock held in London about equal to one year's receipts. This reserve is now about exhausted, and the gum has been steadily advancing in price from the ordinary figures of fourpence to fivepence per pound, until it now commands from one shilling and threepence to two shillings, according to quality.'

ONE BY ONE.

THOUGH from the boughs to which they've long been clinging,
The autumn leaves are dropping one by one,
Yet from their dust, new forms of beauty, springing,
Shall smile again in summer's gentle sun.

Though one by one the pearly drops of morning,
From drooping flowers, on viewless pinions rise,
We'll see them yet the gorgeous clouds adorning
With glowing arches of celestial dyes.

Though one by one the stars are fading slowly
That all night long kept vigil in the sky,
The distant mountain-peaks, like prophets holy,
Proclaim that morning's light and song are nigh.

Though with slow step goes forth the sower weeping,
And on earth's lap his precious treasure leaves,
Yet comes the harvest, with its joyous reaping,
When shall be gathered home the ripened sheaves.

Though one by one the friends we fondly cherish
Withdraw from ours, the cold and trembling hand,
And leave us sorrowful, they do not perish—
They yet shall greet us in a fairer land.

Yes; from all climes, where'er the faithful slumber
'Neath scorching suns, or arctic snow and frost,
Stainless they'll rise, in myriads without number;
All, all, shall meet—there shall not one be lost.

A. M'L.

Printed and Published by W. & R. CHAMBERS, 47 Paternoster Row, LONDON,
and 339 High Street, EDINBURGH.

All Rights Reserved.

*** END OF THE PROJECT GUTENBERG EBOOK CHAMBERS'S
JOURNAL OF POPULAR LITERATURE, SCIENCE, AND ART, FIFTH
SERIES, NO. 40, VOL. I, OCTOBER 4, 1884 ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the

collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with

this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official

version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, “Information about donations to the Project Gutenberg Literary Archive Foundation.”
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.

- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES

EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a)

distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact

links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility: www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.